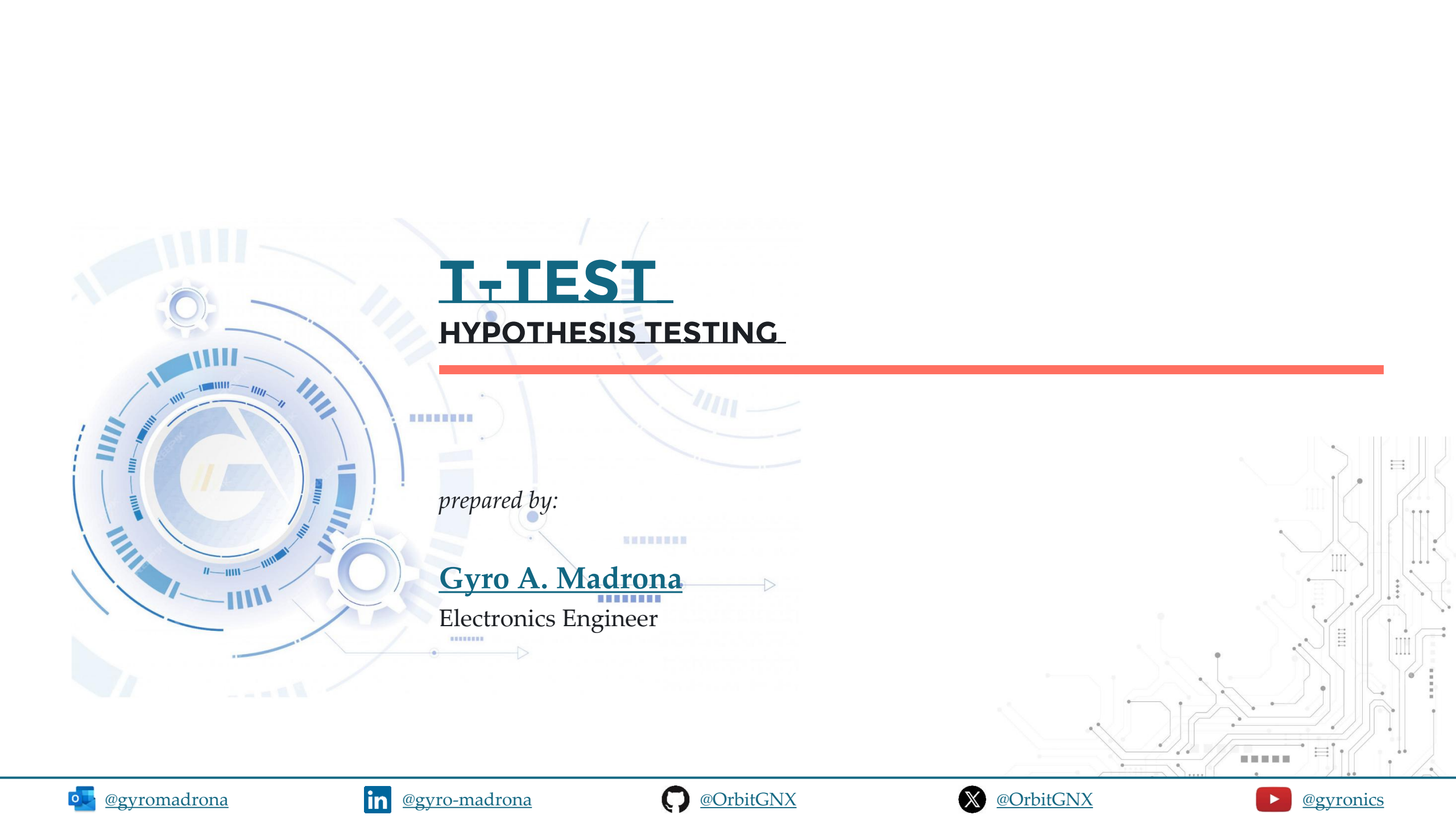




T-TEST

HYPOTHESIS TESTING

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TOPIC OUTLINE

1-Sample t-Test

F-Test

2-Sample t-Test

Paired t-Test



1-SAMPLE T-TEST



1-SAMPLE T-TEST

1-sample t-test is a statistical method used to determine whether the mean of a single sample differs significantly from a known or hypothesized population mean.

Hypothesis

$$H_o: \mu_1 = \mu_o$$

$$H_a: \mu_1 \neq \mu_o \text{ (p-value} < \alpha \text{)}$$

Assumptions

- Continuous data
- Normal data

syntax

```
from scipy import stats

t_stat, p_value = stats.ttest_1samp(
    sample_data, pop_mean)
```

options

```
alternative = 'two-sided' # default
alternative = 'greater'   # u1 > u0
alternative = 'less'      # u1 < u0
```



EXERCISE

The dataset contains the electricity production in MWh by the following production types:

Type	μ	σ
Nuclear	1283.78	32.50
Wind	779.86	88.28
Hydroelectric	1796.86	96.24
Oil and Gas	1160.69	60.66
Coal	1139.33	42.59
Solar	167.58	36.08
Biomass	55.12	2.13

dataset

electricity-normal-sample.csv

Perform a 1-sample t-test to determine whether the mean electricity production (in MWh) for each production type in the given dataset differs significantly from known population parameters.



F-TEST



F-TEST

F-test is a statistical test used to compare the variances of two samples and determine if they are significantly different.

Hypothesis

$$H_o: \sigma_1^2 = \sigma_2^2$$

$$H_a: \sigma_1^2 \neq \sigma_2^2 \text{ (p-value} < \alpha \text{)}$$

Assumptions

- Continuous data
- Normal data

syntax

```
from scipy import stats
```

```
# F-statistic
```

```
if var_1 > var_2:
```

```
    f_stat = var_1/var_2
```

```
else:
```

```
    f_stat = var_2/var_1
```

```
p_value = 1 - stats.f.cdf(
```

```
    f_stat, dof_1, dof_2)
```



2-SAMPLE T-TEST



2-SAMPLE T-TEST

2-sample t-test is a statistical method used to compare the means of two independent groups to determine if they are significantly different from each other.

Hypothesis

$$H_o: \mu_1 = \mu_2$$

$$H_a: \mu_1 \neq \mu_2 \text{ (p-value} < \alpha \text{)}$$

Assumptions

- Continuous data
- Normal data

syntax

```
from scipy import stats

t_stat, p_value = stats.ttest_ind(
    sample_1 data, sample_2 data)
```

options

```
alternative = 'two-sided' # default
alternative = 'greater'   # u1 > u2
alternative = 'less'      # u1 < u2

equal_var = True # default
```



EXERCISE

The dataset contains the electricity production in MWh by the following production types:

Type	μ	σ
Nuclear	1283.78	32.50
Wind	779.86	88.28
Hydroelectric	1796.86	96.24
Oil and Gas	1160.69	60.66
Coal	1139.33	42.59
Solar	167.58	36.08
Biomass	55.12	2.13

dataset

electricity-normal-sample.csv

Perform a 2-sample t-test to compare the means of electricity production (in MWh) between two distinct groups in the dataset.



PAIRED T-TEST



PAIRED T-TEST

Paired t-test is a statistical method used to compare the means of two related groups to determine if they are significantly different from each other.

Hypothesis

$$H_0: \mu_1 = \mu_2$$

$$H_a: \mu_1 \neq \mu_2 \quad (\text{p-value} < \alpha)$$

Assumptions

- Continuous data
- Normal data

syntax

```
from scipy import stats

t_stat, p_value = stats.ttest_rel(
    sample_1_data, sample_2_data)
```

options

```
alternative = 'two-sided' # default
alternative = 'greater'   # u1 > u2
alternative = 'less'      # u1 < u2

equal_var = True # default
```



EXERCISE

The dataset records the nuclear power electricity production (in MWh) from the same reactors before and after maintenance. Perform a paired t-test to determine whether the maintenance is effective.

dataset

nuclear-maintenance.csv

Solution



LABORATORY

